Detailed Design:

SNo	Module	Coverage	Duration
1	**Quick overview of Generative AI**	• What is Generative AI vs Predictive AI • How LLMs generate text (tokens, probability, decoding) • Capabilities & limitations of LLMs • Hallucinations & why retrieval is needed • Role of embeddings and RAG in grounding GenAI	1
2	**Vectorization and Similarity techniques**	• Transformer-based word, sentence & document embeddings • Converting text into vector representations • Similarity techniques (cosine, dot-product, semantic similarity) • **Hands-on**: Generate embeddings & compare text similarity	2
3	**LangChain**	• Introduction to LangChain Framework • Chat models & prompt templates • Chains & LCEL (LangChain Expression Language) patterns • Memory (buffer, summary, vector memory) • Tools & function calling • Agents (ReAct, planner-executor) • Output parsers for structured responses	5
4	**Vector Databases + RAG**	• Overview of RAG • Core components • Common use cases • Embeddings and Semantic search • Chunking strategies and best practices • Vector Databases: concept, types • How to upload data into a vector database • Query a vector database, generating the answer & concatenating the relevant documents • Visualizing embedding spaces for similarity validation & debugging • **Hands-on**: Build a QA application using a policy document PDF	6

5	**RAG Evaluation Techniques**	• Retrieval Quality (Intrinsic evaluation)
		  ○ Recall@k, Precision@k, MAP (Mean Average Precision), MRR (Mean Reciprocal Rank) etc.
		• End-to-End QA (Extrinsic evaluation)
		  ○ Exact match, F1, ROUGE/BLEU etc.
		• LLM-as-a-Judge technique
			2

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